Laplace transform of periodic functions

1. Find Laplace transform of $f(t)=\begin{cases} t, 0<t<\pi \\ \pi-t, \pi<t<2\pi \end{cases}$

   ans: $\dfrac{1}{1-e^{-2\pi s}}\left\{\dfrac{\pi}{s}\left(e^{-2\pi s}-e^{-\pi s}\right)+\dfrac{1}{s^2}\left(1+e^{-2\pi s}-2e^{-\pi s}\right)\right\}$

2. Find the Laplace transform of the square wave function of period 2a defined by $f(t)=\begin{cases} k, 0<t<a \\ k, 0<t<2a \end{cases}$

   Ans: $\dfrac{k}{s}\tanh\left(\dfrac{as}{2}\right)$

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